



OSCILLATIONS, WAVES AND OPTICS

PHYSICS FOR CHEMISTRY STUDENTS

Ye Yang
Xiamen University

REVIEW

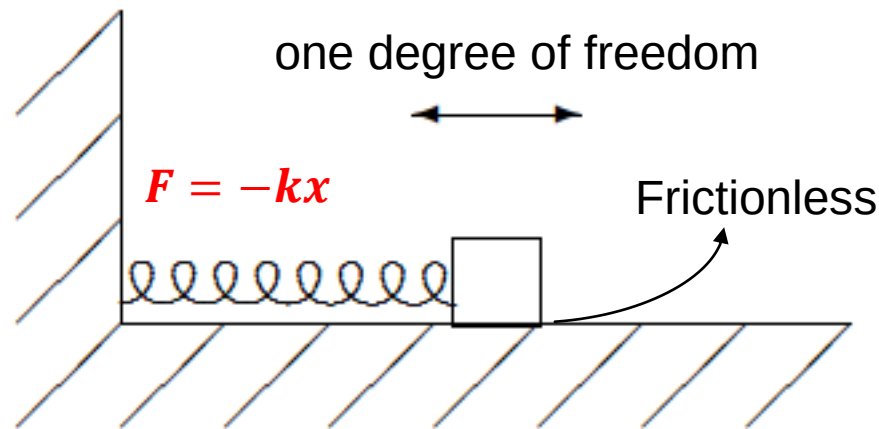


Figure 1.1: A mass on a spring.

Equation of motion: $m\ddot{x}(t) = -kx(t)$

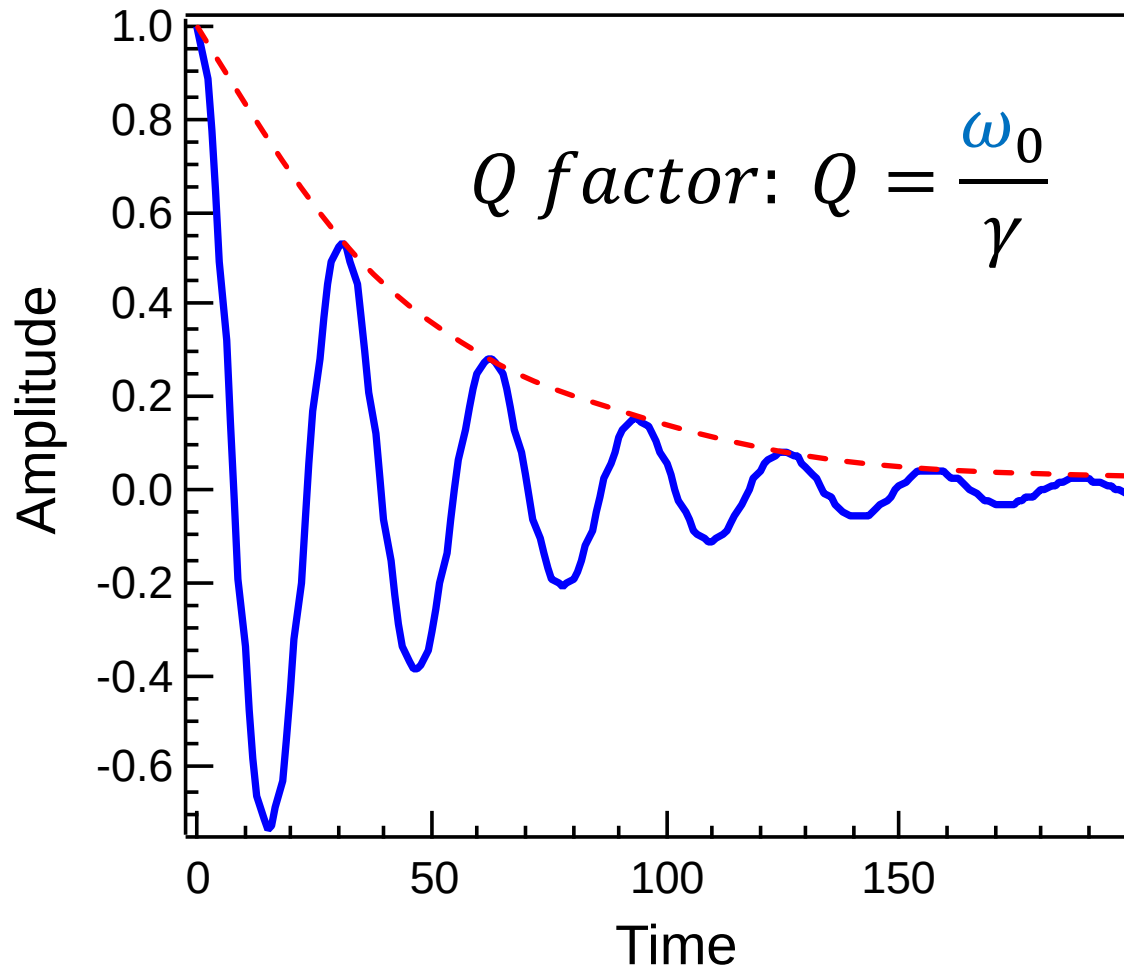
Solution: $x(t) = A \cos(\omega t + \phi)$

Energy conservation: $\partial \left(\frac{1}{2} m \dot{x}^2(t) + V(x) \right) / \partial t = 0$

REVIEW

$$\ddot{x} + \gamma \dot{x} + \omega_0^2 x(t) = 0$$

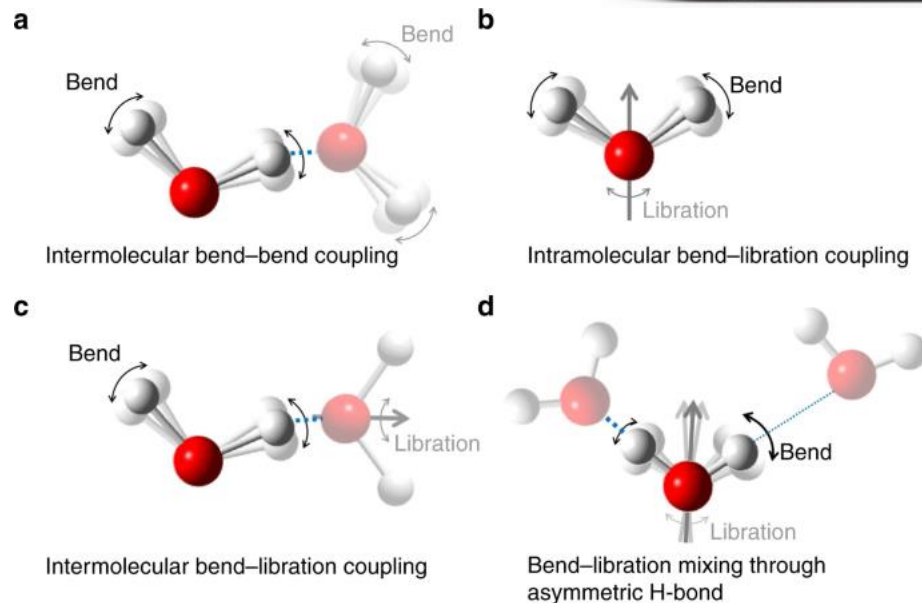
$$x(t) = Ae^{-\frac{\gamma}{2}t} \cos(\omega t + \phi)$$



CHAPTER 1. OSCILLATION

1.3. DRIVEN OSCILLATOR

External driving force ?



1.3. DRIVEN OSCILLATOR

Hooke's Law

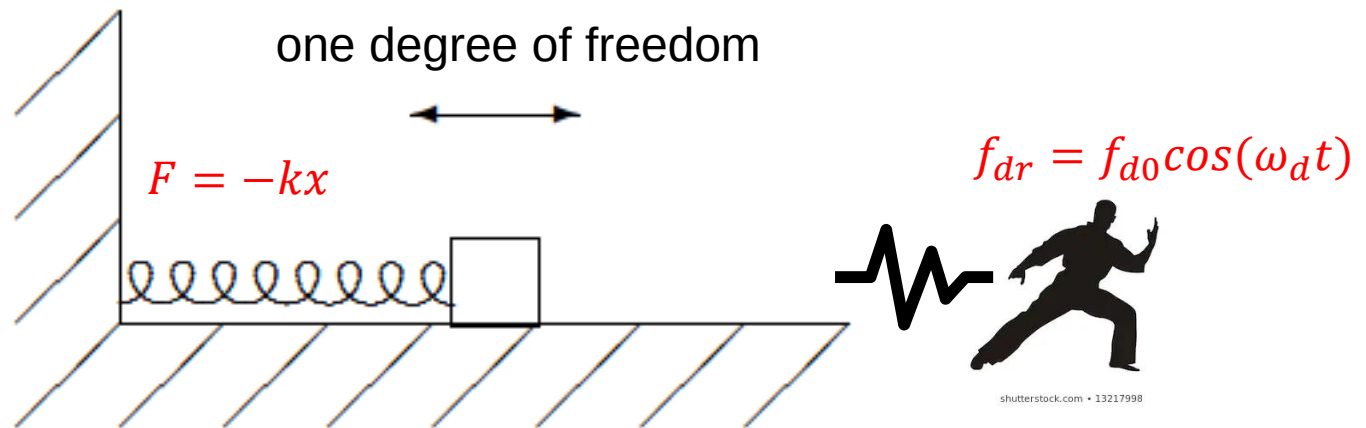


Figure 1.1: A mass on a spring.

Equation of motion

$$m\ddot{x} = -kx + f_{d0}\cos(\omega_d t)$$

1.3. DRIVEN OSCILLATOR

Equation of motion

$$\ddot{x}(t) = -\omega_0^2 x(t) + f_0 \cos(\omega_d t)$$

$$\omega_0 = \sqrt{k/m}$$
$$f_0 = \frac{f_{d0}}{m}$$

General solution

$$x(t) = x_c(t) + x_p(t)$$

General solution of the
corresponding homogeneous equation

$$x_c(t) = A \sin(\omega_0 t + \phi)$$

1.3. DRIVEN OSCILLATOR

Equation of motion

$$\ddot{x}(t) = -\omega_0^2 x(t) + f_0 \cos(\omega_d t)$$

General solution

$$x(t) = x_c(t) + x_p(t)$$

Particular solution

Auxiliary equation

$$\ddot{x}(t) = -\omega_0^2 x(t) + f_0 \cos(\omega_d t)$$

$$\omega_0 \neq \omega_d$$

Trial solution

$$x(t) = B \cos(\omega_d t)$$

$$-\omega_d^2 B \cos(\omega_d t) = -\omega_0^2 B \cos(\omega_d t) + f_0 \cos(\omega_d t)$$

1.3. DRIVEN OSCILLATOR

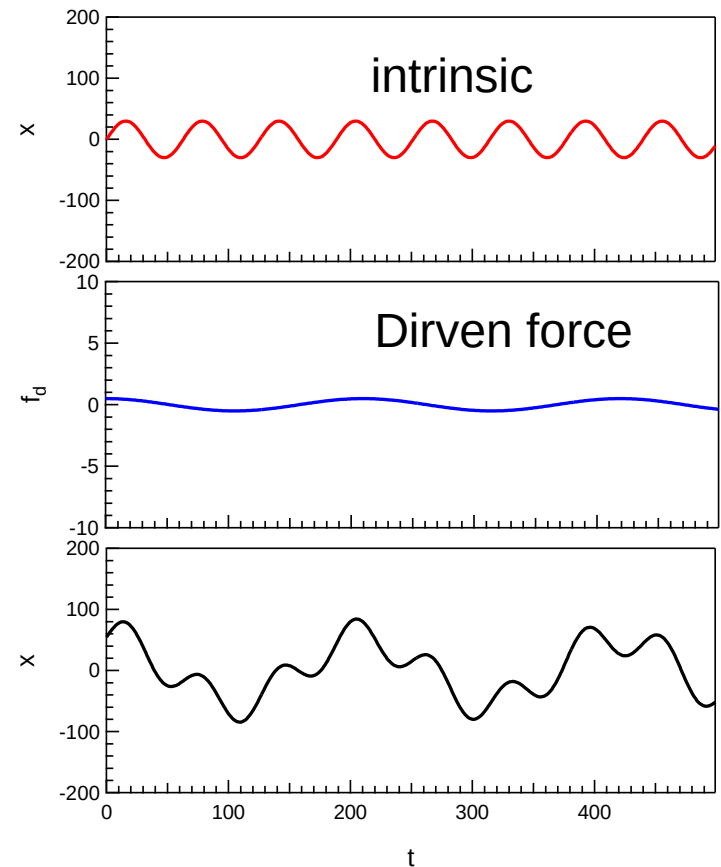
$$(-\omega_d^2 + \omega_0^2)B = f_0$$

$$B = \frac{f_0}{(-\omega_d^2 + \omega_0^2)}$$

$$x(t) = x_c(t) + x_p(t)$$

$$x_p(t) = B \cos(\omega_d t)$$

$$x(t) = A \sin(\omega_0 t + \phi) + \frac{f_0}{(-\omega_d^2 + \omega_0^2)} \cos(\omega_d t)$$



$$\omega_d = 0.3\omega_0$$

1.3. DRIVEN OSCILLATOR

$$(-\omega_d^2 + \omega_0^2)B = f_0$$

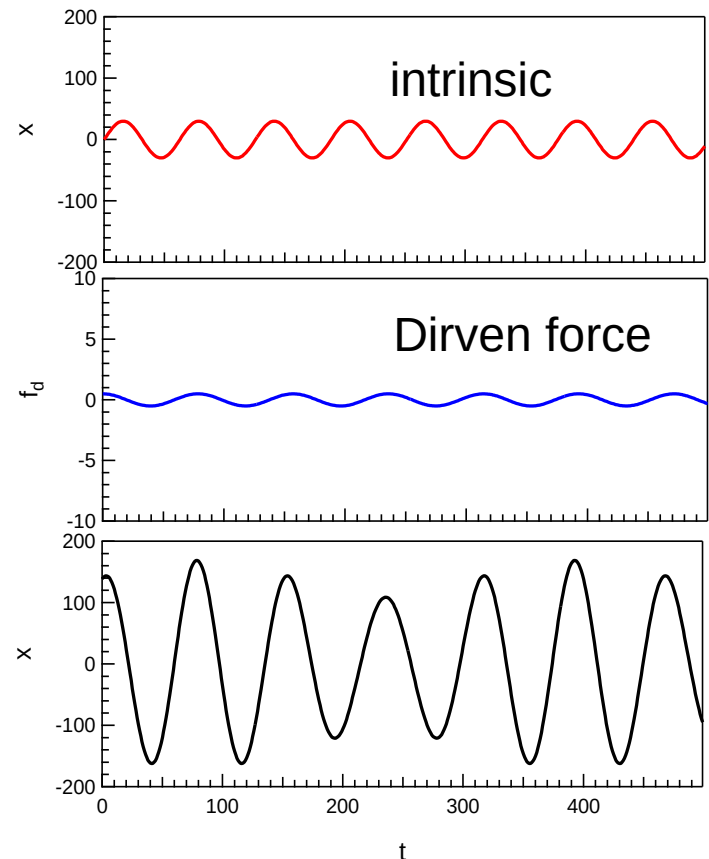
$$B = \frac{f_0}{(-\omega_d^2 + \omega_0^2)}$$

$$x(t) = x_c(t) + x_p(t)$$

$$x_p(t) = B \cos(\omega_d t)$$

$$x(t) = A \sin(\omega_0 t + \phi) + \frac{f_0}{(-\omega_d^2 + \omega_0^2)} \cos(\omega_d t)$$

$$\omega_d \rightarrow \omega_0 \quad ?$$



$$\omega_d = 0.8\omega_0$$

1.3. DRIVEN OSCILLATOR

Equation of motion

$$\ddot{x}(t) = -\omega_0^2 x(t) + f_0 \cos(\omega_d t)$$

$$\omega = \sqrt{k/m}$$

General solution

$$x(t) = x_c(t) + x_p(t)$$

Particular solution

Auxiliary equation

$$\ddot{x}(t) = -\omega_0^2 x(t) + f_0 e^{i\omega_d t}$$

$$\omega_0 = \omega_d$$

Trial solution

$$x(t) = B t e^{i\omega_d t}$$

$$i2\omega_d B e^{i\omega_d t} - \omega_d^2 B t e^{i\omega_d t} = -\omega_0^2 B t e^{i\omega_d t} + f_0 e^{i\omega_d t}$$

1.3. DRIVEN OSCILLATOR

$$i2\omega_d B e^{i\omega_d t} = f_0 e^{i\omega_d t}$$

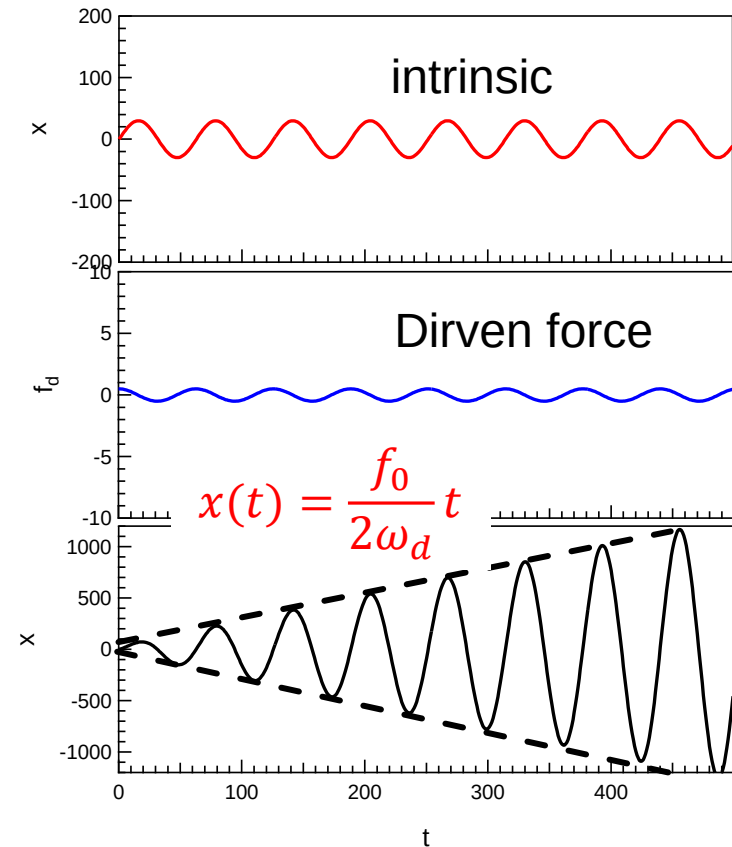
$$B = \frac{f_0}{i2\omega_d}$$

$$x(t) = x_c(t) + x_p(t)$$

$$x_p(t) = B t e^{i\omega_d t}$$

$$x(t) = A \sin(\omega_0 t + \phi) + \frac{f_0}{2\omega_d} t \sin(\omega_d t)$$

Resonance



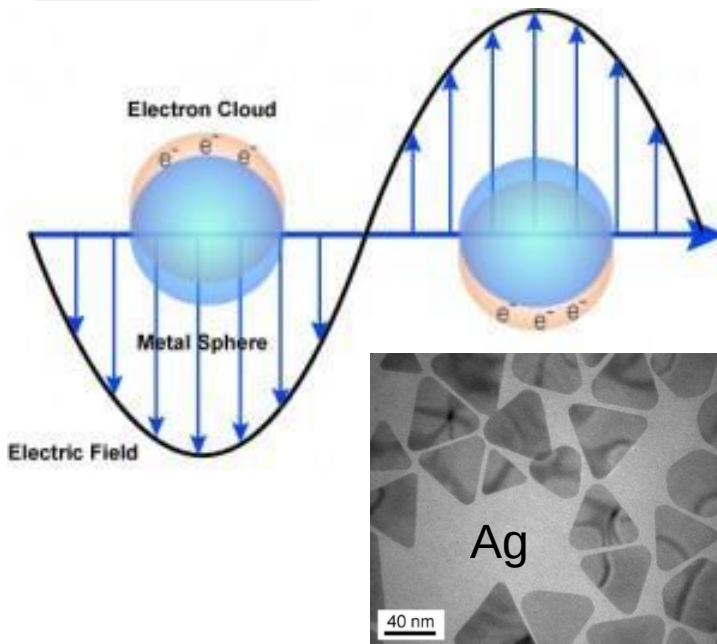
1.3. DRIVEN OSCILLATOR

Resonance

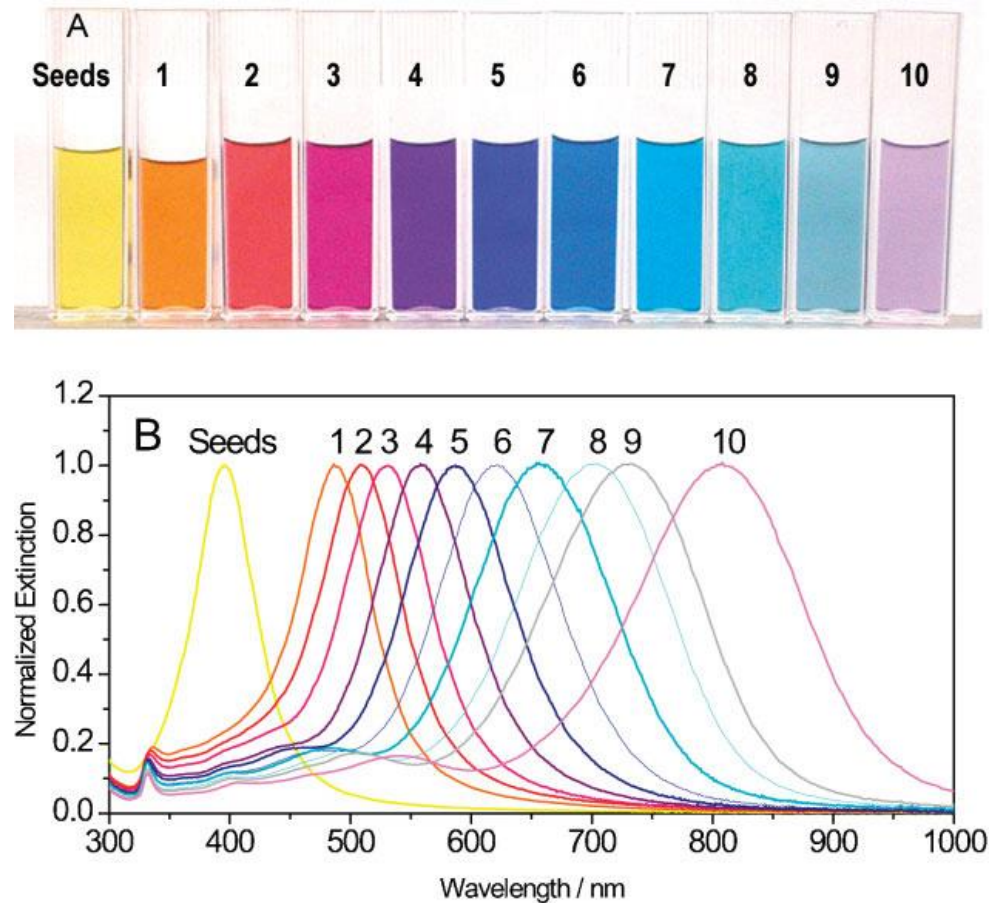


1.3. DRIVEN OSCILLATOR

Resonance



Local Surface Plasmon Resonance



1.3. DRIVEN OSCILLATOR

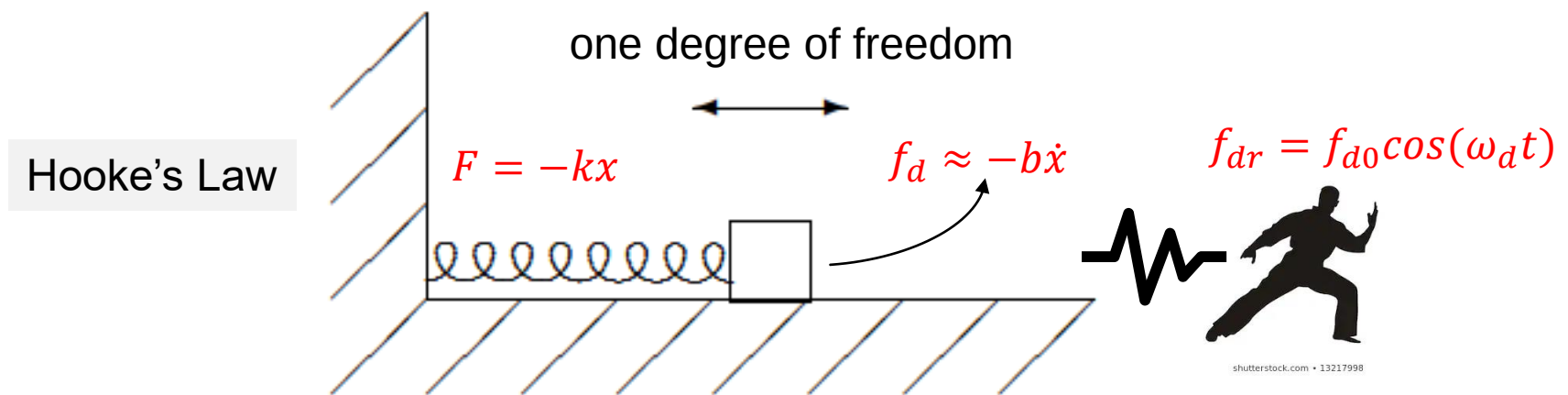


Figure 1.1: A mass on a spring.

Equation of motion

$$m\ddot{x} = -kx - b\dot{x} + f_{d0}\cos(\omega_d t)$$

1.3. DRIVEN OSCILLATOR

Equation of motion

$$\ddot{x}(t) = -\omega_0^2 x(t) - \gamma \dot{x}(t) + f_0 \cos(\omega_d t)$$

$$\omega_0 = \sqrt{k/m}$$
$$\gamma = b/m$$

General solution

$$x(t) = x_c(t) + x_p(t)$$

Particular solution

General solution of the
corresponding homogeneous equation

Complex expression

$$\ddot{x}(t) = -\omega_0^2 x(t) - \gamma \dot{x}(t) + f_0 e^{i\omega_d t}$$

Trial solution

$$x(t) = B e^{i\omega_d t}$$

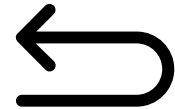
1.3. DRIVEN OSCILLATOR

Equation of motion

$$\ddot{x}(t) + \gamma \dot{x}(t) + \omega_0^2 x(t) = f_0 e^{i\omega_d t}$$

Trial solution

$$x(t) = B e^{i\omega_d t}$$



$$\ddot{x}(t) = -\omega_d^2 B e^{i\omega_d t} \quad \dot{x}(t) = i\omega_d B e^{i\omega_d t}$$



$$(-\omega_d^2 + i\omega_d \gamma + \omega_0^2) B = f_0$$

If B is real

Real part:

$$(-\omega_d^2 + \omega_0^2) B = f_0$$

Imag part:

$$(\omega_d \gamma) B = 0$$

$$\gamma = 0$$

1.3. DRIVEN OSCILLATOR

$$(-\omega_d^2 + i\omega_d\gamma + \omega_0^2)B = f_0$$

If B is complex

$$(-\omega_d^2 + i\omega_d\gamma + \omega_0^2)B_0 e^{-i\delta} = f_0$$

Real part:

$$(-\omega_d^2 + \omega_0^2)B_0 = f_0 \cos(\delta)$$

Imag part:

$$(\omega_d\gamma)B_0 = f_0 \sin(\delta)$$

$$B_0 = \frac{f_0}{\sqrt{(\omega_0^2 - \omega_d^2)^2 + (\omega_d\gamma)^2}}$$

$$\tan(\delta) = \frac{\omega_d\gamma}{\omega_0^2 - \omega_d^2}$$

1.3. DRIVEN OSCILLATOR

Equation of motion

$$\ddot{x}(t) + \gamma \dot{x}(t) + \omega_0^2 x(t) = f_0 e^{i\omega_d t}$$

Particular solution

$$x(t) = B_0 e^{-i\delta} e^{i\omega_d t}$$

$$B_0 = \frac{f_0}{\sqrt{(\omega_0^2 - \omega_d^2)^2 + (\omega_d \gamma)^2}} \quad \text{tg}(\delta) = \frac{\omega_d \gamma}{\omega_0^2 - \omega_d^2}$$

General solution

Phase delay of the response

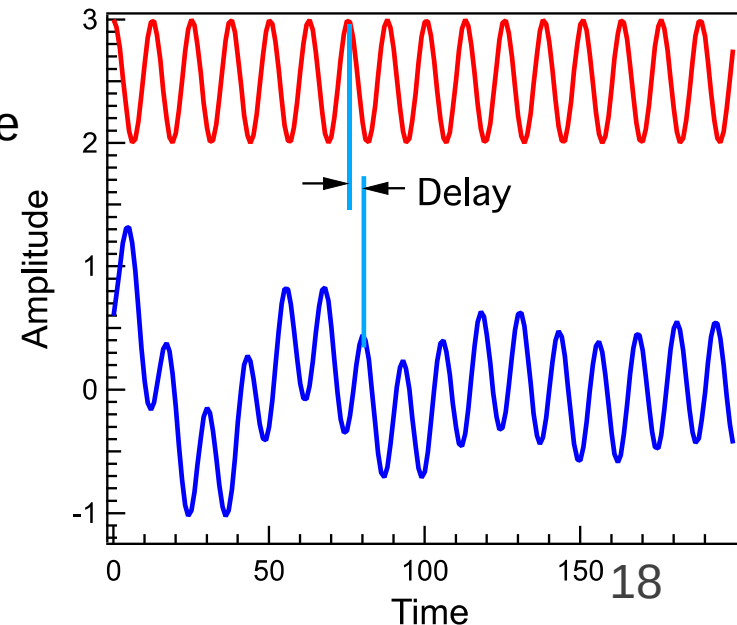
$$x(t) = x_c(t) + x_p(t)$$

$$x(t) = Ae^{-\frac{\gamma}{2}t} \sin(\omega t + \phi) + B_0 e^{i\omega_d(t - \delta/\omega_d)}$$

Die out

Steady state

underdamped



1.4. Q FACTOR & RESONANCE

Equation of motion

$$m\ddot{x} = -kx - b\dot{x} + f_{d0}\cos(\omega_d t)$$

Particular solution

$$x(t) = B_0 e^{i(\omega_d t - \delta)}$$

$$B_0 = \frac{f_0}{\sqrt{(\omega_0^2 - \omega_d^2)^2 + (\omega_d \gamma)^2}} \quad \text{tg}(\delta) = \frac{\omega_d \gamma}{\omega_0^2 - \omega_d^2}$$

$$\omega_d \rightarrow \omega_0; Q = \frac{\omega_0}{\gamma} > 1/\sqrt{2}$$

$$B_0 \rightarrow B_{\max}$$

$$\begin{aligned} \text{tg}(\delta) &\rightarrow \infty \\ \delta &\rightarrow \frac{\pi}{2} \end{aligned}$$

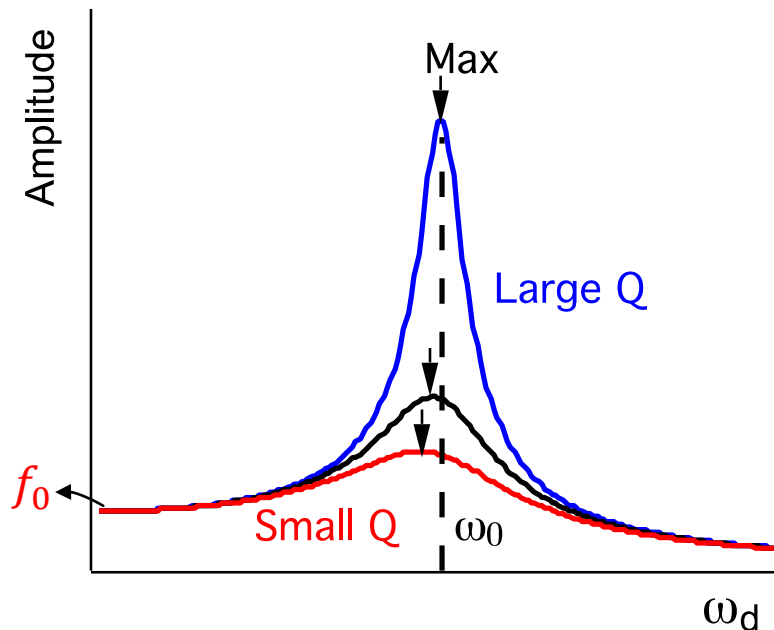
1.4. Q FACTOR & RESONANCE

Resonance vs Q factor

$$\omega_d \rightarrow \omega_0$$

$$B_0 = \frac{f_0}{(\omega_d \gamma)} = \frac{f_0}{\omega_0^2} \frac{\omega_0}{\gamma} = \frac{f_{d0}}{k} Q$$

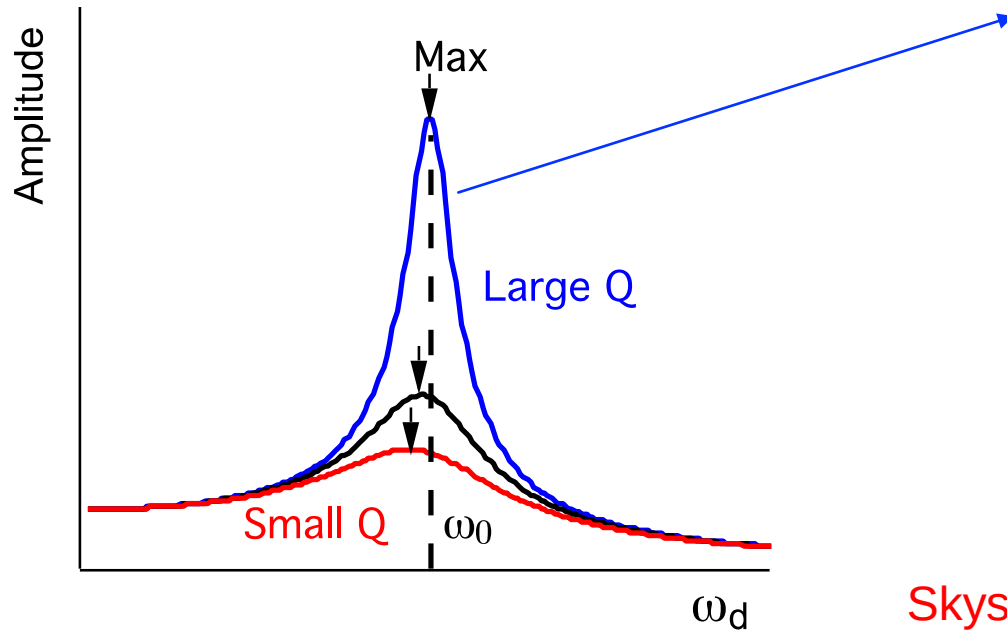
Dimensionless



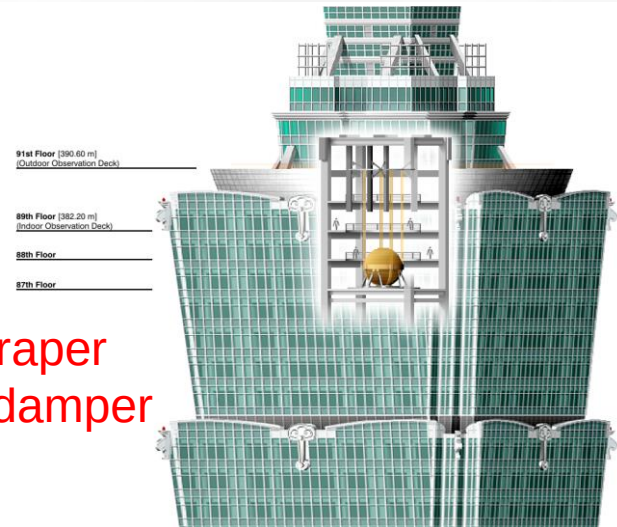
$$Q = \frac{\omega_0}{\gamma}$$

$$B_0 = \frac{Q f_0}{\sqrt{Q^2 (1 - \omega_d^2 / \omega_0^2)^2 + (\omega_d / \omega_0)^2}}$$

1.4. Q FACTOR & RESONANCE



Skyscraper
Wind damper



CHAPTER 1. OSCILLATION

time-resolved terahertz spectroscopy



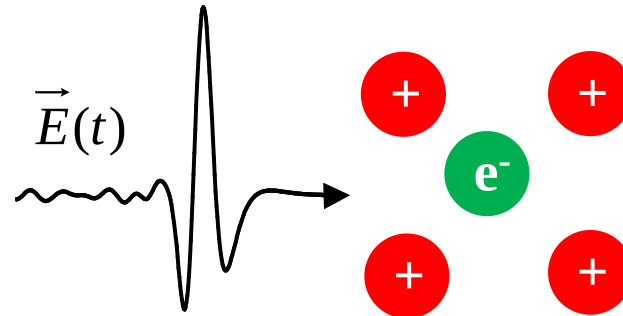
CASE STUDY

Measure photoconductivity using THz spectroscopy

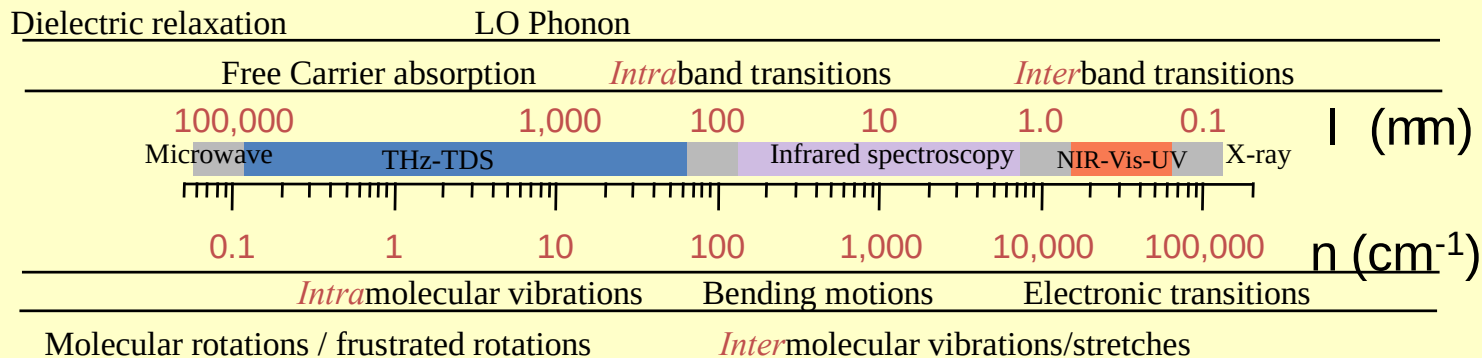
$$\mathbf{j} = E\sigma$$

$$\mathbf{j} = \lim_{\Delta t \rightarrow 0} \frac{\Delta Q}{\mathbf{S} \cdot \Delta \mathbf{t}} = \lim_{\Delta t \rightarrow 0} \frac{-nVq}{\mathbf{S} \cdot \Delta \mathbf{t}} = -nq\dot{\mathbf{x}}$$

$$\sigma = -nq\dot{\mathbf{x}}/E$$



• 1 THz = 33 cm⁻¹ = 4 meV = 300 μm

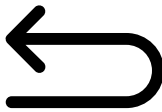


CHAPTER 1. OSCILLATION

1. Force analysis (Physics)

$$F_{total} = F_k + F_d + F_f$$

$$\ddot{x} + \gamma\dot{x} + \omega_0^2 x = -\frac{q}{m} E_0 e^{-i\omega_d t}$$



Trial solution

$$x(t) = A e^{-i\omega_d t}$$

restoring $F_k = -kx$ (*small oscillation*)

driving $F_d = -qE = -qE_0 e^{-i\omega_d t}$

damping $F_s = -m\gamma\dot{x}$

$$(-\omega_d^2 - i\omega_d\gamma + \omega_0^2)A = -\frac{q}{m} E_0$$

~~$$x(t) = B e^{-\gamma t} e^{i\omega t} + A e^{-i\omega_d t}$$~~

2. Equation of motion (Math)

$$m\ddot{x} = -kx - qE_0 e^{-i\omega_d t} - m\gamma\dot{x}$$

Initial condition: electron is at the equilibrium position

$$\dot{x}(t) = -i\omega_d A e^{-i\omega_d t}$$

CHAPTER 1. OSCILLATION

$$\dot{x}(t) = -i\omega_d A e^{-i\omega_d t}$$

$$\sigma = -nq\dot{x}/E$$

$$E = E_0 e^{-i\omega_d t}$$

$$\sigma = nqi\omega_d A e^{-i\omega_d t} / E_0 e^{-i\omega_d t}$$

$$(-\omega_d^2 - i\omega_d\gamma + \omega_0^2)A = -\frac{q}{m}E_0$$

$$\sigma = nqi\omega_d A / E_0$$

$$A/E_0 = -\frac{q}{m} / (-\omega_d^2 - i\omega_d\gamma + \omega_0^2)$$

$$\sigma = -nqi\omega_d \cdot \frac{q}{m} / (-\omega_d^2 - i\omega_d\gamma + \omega_0^2)$$

$$\sigma = -\frac{i\omega_d \epsilon_0 \omega_p^2}{(\omega_0^2 - \omega_d^2 - i\omega_d\gamma)}$$

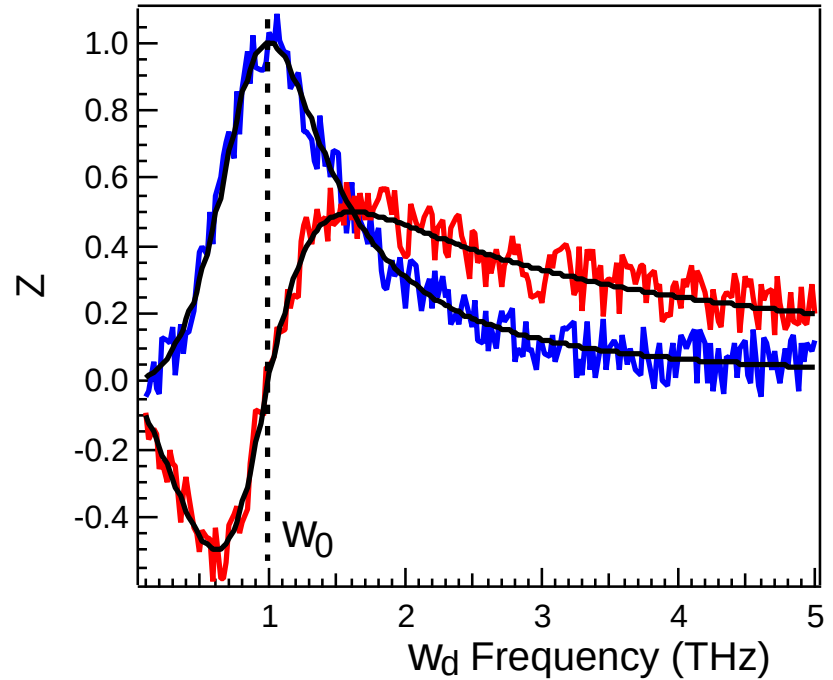
$$\omega_p^2 = \frac{nq^2}{m\epsilon_0}$$

CHAPTER 1. OSCILLATION

$$\sigma = -\frac{i\omega_d \epsilon_0 \omega_p^2}{(\omega_0^2 - \omega_d^2 - i\omega_d \gamma)}$$

Electron mobility:

$$\mu = -\frac{q}{m\gamma}$$



CHAPTER 1. OSCILLATION

Free electron model

$$V(x) = V_0 \quad F_k(x) = -\frac{\partial V}{\partial x} = 0$$

1. Force analysis (Physics)

$$F_{total} = \cancel{F_k} + F_d + F_s$$

$$\ddot{x} + \gamma\dot{x} = -\frac{q}{m}E_0 e^{-i\omega_d t}$$

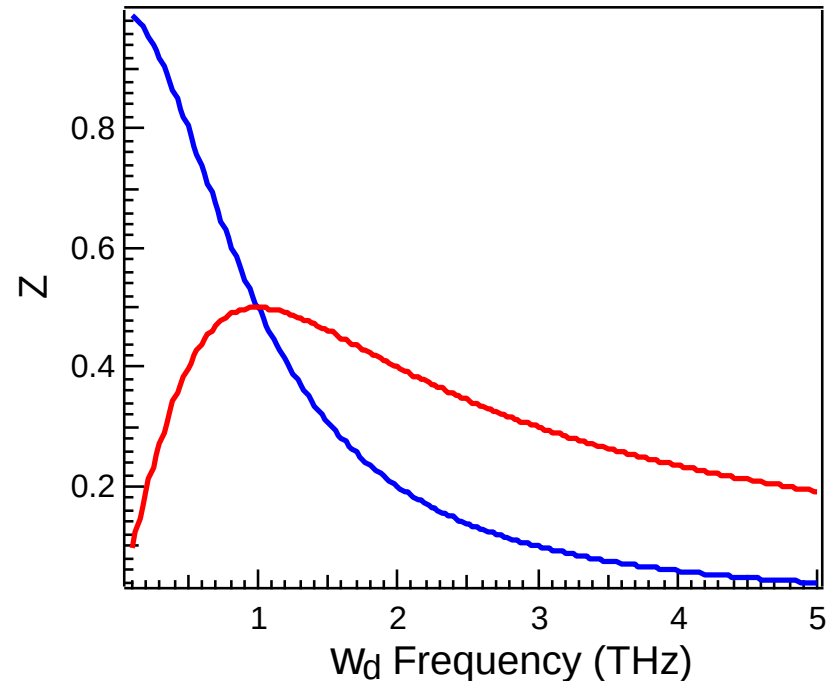
$$(-\omega_d^2 - i\omega_d\gamma)A = -\frac{q}{m}E_0$$

Drude model

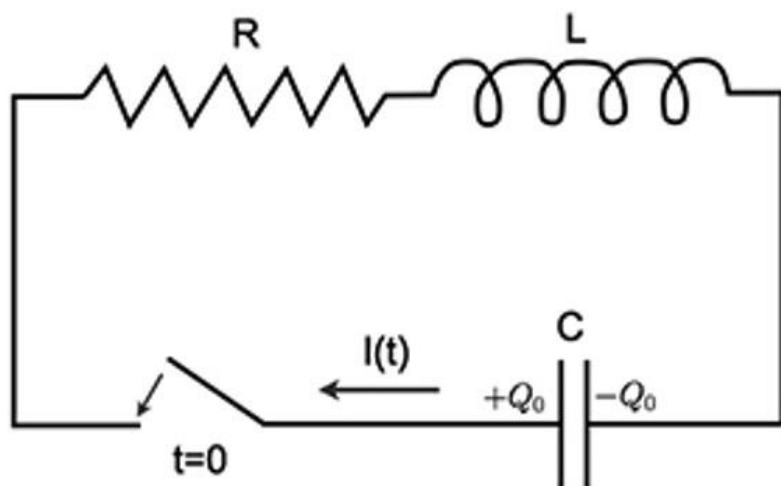
$$\sigma = -\frac{i\varepsilon_0\omega_p^2}{(\omega_d + i\gamma)}$$

Electron mobility:

$$\mu = -\frac{q}{m\gamma}$$



HOMEWORK



如图所示，某RLC电路由电容(C)、电阻(R)、电感(L)各一个组成，其中电容储存电荷容量为 Q_0 ，电路开关处于打开状态（电路未接通）。假设闭合开关（接通电路）的时刻设为时间零点。

1. R 的取值处于什么范围时，该RLC电路会发生多次振荡。
2. 假设 R 处在上述取值范围，写出电路中的电流以及电容中的电荷随时间变化的函数，即 $I(t)$ ， $Q(t)$ 。
3. 如果电路中的储存的能量衰减至起始状态的 $1/e$ （ e 为自然常数）时电路发生了10次振荡，写出电阻 R 与电感 L 和电容 C 的关系表达式。
4. 写出该电路品质因子表达式。

$$(1) V(t) = V_L(t) + V_C(t) + V_R(t) = 0$$

$$V_R(t) = R \cdot I(t)$$

$$V_L(t) = L \cdot \frac{dI(t)}{dt}$$

$$V_C(t) = \int I(t) dt / C$$

$$R \cdot I(t) + L \cdot \frac{dI(t)}{dt} + \frac{\int I(t) dt}{C} = 0$$

$$\frac{d^2 I(t)}{dt^2} + \frac{R}{L} \frac{dI(t)}{dt} + \frac{1}{LC} I(t) = 0$$

$$r = \frac{R}{L} \quad \omega_0^2 = \frac{1}{LC} \quad \ddot{I} + r\dot{I} + \omega_0^2 I = 0$$

$$r^2 - 4\omega_0^2 < 0 \text{ if } \frac{R^2}{L^2} - \frac{4}{LC} < 0 \quad R < 2\sqrt{\frac{L}{C}}$$

$$(2) s^2 + 2\alpha s + \omega_0^2 = 0$$

$$s_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2} \quad s_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2}$$

$$I(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

$$\zeta = \frac{\alpha}{\omega_0} < 1 \quad I(t) = B_1 \cos \omega_d t + i B_2 \sin \omega_d t$$

$$(\omega_d = \sqrt{\omega_0^2 - \alpha^2})$$

$$Q(t) = Q_0 - \int_0^t I(t) dt = Q_0 - (2B_1 \sin \omega_d t - i \omega_d B_2 \cos \omega_d t + i \omega_d B_2)$$

$$(3) T = \frac{2\pi}{\omega} \quad t = 10T = \frac{20\pi}{\omega}$$

$$E = \frac{1}{2} \frac{Q^2}{C} + \frac{1}{2} L \dot{I}^2 = \frac{1}{2} \frac{Q^2}{C} = \dots$$

$$Q = \frac{1}{\sqrt{e}} Q_0$$

$$Q_0 - \frac{C}{1 - r/2} = \frac{1}{\sqrt{e}} Q_0$$

$$(4) Q = \frac{\sqrt{1/LC}}{R/L} = \frac{1}{R} \sqrt{\frac{L}{C}}$$

$$\ddot{I} + r\dot{I} + \omega^2 I = 0 \quad \wedge \quad r^2 - 4\omega^2 = -4\omega^2$$

$$x = \frac{-r \pm i2\omega}{2} t$$

$$I(t) = e^{-\frac{r}{2}t} (a \sin \omega t + b \cos \omega t) = A e^{-\frac{r}{2}t} \sin(\omega t + \varphi) \quad I(0) = A \sin \varphi = 0 \quad \varphi = 0$$

$$Q(t) = Q_0 - \int_0^t I(t) dt = Q_0 - \int_0^t A e^{-\frac{r}{2}t} \sin \omega t dt = Q_0 - \frac{-\frac{2A}{r} e^{-\frac{r}{2}t} \sin \omega t - \frac{4A}{r^2} (e^{-\frac{r}{2}t} \cos \omega t - 1)}{\frac{4}{r^2} + 1}$$

$$Q\left(\frac{20\pi}{\omega}\right) = Q_0 + \frac{4A(e^{-\frac{10\pi r}{\omega}} - 1)}{4 + r^2} = \frac{1}{\sqrt{e}} Q_0$$

$$(1 - \frac{1}{\sqrt{e}}) Q_0 = \frac{4A(1 - e^{-\frac{20\pi r}{\omega}})}{4 + \frac{r^2}{\omega^2}} \quad \frac{\sqrt{4\omega^2 - r^2}}{2}$$

$$-\frac{2A}{r} e^{-\frac{r}{2}t} \sin(\omega t) + \int \frac{2A}{r} e^{-\frac{r}{2}t} \cos(\omega t) dt = \int A e^{-\frac{r}{2}t} \sin(\omega t) dt$$

$$\int_0^t e^{-\frac{r}{2}t} \cos(\omega t) dt = -\frac{2}{r} e^{-\frac{r}{2}t} \sin(\omega t) \Big|_0^t = \int_0^t \frac{2}{r} e^{-\frac{r}{2}t} \sin(\omega t) dt$$

$$\int_0^t e^{-\frac{r}{2}t} \cos(\omega t) dt = \left(\frac{2}{r} e^{-\frac{r}{2}t} \sin(\omega t) \right) \Big|_0^t - \int_0^t \frac{2}{r} e^{-\frac{r}{2}t} \sin(\omega t) dt$$

$$\left(-\frac{2A}{r} e^{-\frac{r}{2}t} \sin \omega t \right) \Big|_0^t + \frac{2A}{r} \left(-\frac{2}{r} e^{-\frac{r}{2}t} \cos(\omega t) \right) \Big|_0^t - \int_0^t \frac{2}{r} e^{-\frac{r}{2}t} \sin \omega t dt = \int_0^t A e^{-\frac{r}{2}t} \sin(\omega t) dt$$

$$-\frac{2A}{r} e^{-\frac{r}{2}t} \sin \omega t - \frac{4A}{r^2} (e^{-\frac{r}{2}t} \cos \omega t - 1) - \frac{4A}{r^2} \int_0^t e^{-\frac{r}{2}t} \sin \omega t dt = A \int_0^t e^{-\frac{r}{2}t} \sin \omega t dt$$